

Question 2

Max. score: 100.00

1

Alice throws a party

2

Alice is going to throw a party for some of her friends. At the party, Alice would give each of her friends exactly X candies but the constraint is that for every friend of hers, the X candies given to him must be of a different flavor, i.e., there must be X distinct flavored candies given to each of her friends. Although two different friends can have the same flavored candy. Now she has an array A of length N where $A[i]$ represents the number of candies of type i available to her.

Task Determine the maximum number of friends she can invite to the party satisfying the given conditions.

Notes

- The answer may not fit in a 32-bit integer.
- 1-based indexing is followed.

Example

Assumptions

- $N = 3$
- $X = 2$

New Submission

All Submissions

① Auto-complete connection failed.
Reconnect

C++17 (g++ 10.3.0)



```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 long long solve (int N, int X, vector<int> A) {
5     // code starts here
6     sort(A.begin(),A.end());
7     long long ans = 0;
8     priority_queue<int,vector<int>,greater<int>>pq;
9     for(int i=0; i<N; i++)
10         pq.push(A[i]);
11
12 int main() {
13
14     ios::sync_with_stdio(0);
15     cin.tie(0);
16     int T;
17     cin >> T;
18     for(int t_i = 0; t_i < T; t_i++)
19     {
20         int N;
21         cin >> N;
22         int X;
23         cin >> X;
24         vector<int> A(N);
```

9:12 vscode

Input

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Compile & Test code

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invite to the party satisfying the given conditions.

Notes

- The answer may not fit in a 32-bit integer.
- 1-based indexing is followed.

Example

Assumptions

- $N = 3$
- $X = 2$
- $A = [1, 4, 2]$

Approach

- You have to give exactly 2 candies to each of the friends satisfying that these 2 must be distinct for each friend.
- You can invite at max 3 friends by giving them candies as follows (array of each friend contains the type of candy)
 - Friend 1 gets candies of type (all distinct) $\Rightarrow [1, 2]$
 - Friend 2 gets candies of type (all distinct) $\Rightarrow [2, 3]$
 - Friend 3 gets candies of type (all distinct) $\Rightarrow [2, 3]$
- Thus the answer is 3.

Function description

New Submission

All Submissions

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C++17 (g++ 10.3.0)



```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 long long solve (int N, int X, vector<int> A) {
5     // code starts here
6     sort(A.begin(),A.end());
7     long long ans = 0;
8     priority_queue<int,vector<int>,greater<int>>pq;
9     for(int )
10 }
11
12 int main() {
13
14     ios::sync_with_stdio(0);
15     cin.tie(0);
16     int T;
17     cin >> T;
18     for(int t_i = 0; t_i < T; t_i++)
19     {
20         int N;
21         cin >> N;
22         int X;
23         cin >> X;
24         vector<int> A(N);
```

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Test against custom input ▼

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Compile & Test code

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1

2

- You have to give exactly 2 candies to each of the friends satisfying that these 2 must be distinct for each friend.
- You can invite at max 3 friends by giving them candies as follows (array of each friend contains the type of candy)
 - Friend 1 gets candies of type (all distinct) => [1, 2]
 - Friend 2 gets candies of type (all distinct) => [2, 3]
 - Friend 3 gets candies of type (all distinct) => [2, 3]
- Thus the answer is 3.

Function description

Complete the function `solve` provided in the editor. This function takes the following 3 parameters and returns the required answer:

- `N`: Represents the size of array `A`
- `X`: Represents the number of candies to be given to each friend
- `A`: Represents the numbers of candies of each type

Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button).

- The first line contains `T` denoting the number of test

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All Submissions

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C++17 (g++ 10.3.0)



```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 long long solve (int N, int X, vector<int> A) {
5     // code starts here
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7     long long ans = 0;
8     priority_queue<int,vector<int>,greater<int>>pq;
9     for(int )
10 }
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12 int main() {
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14     ios::sync_with_stdio(0);
15     cin.tie(0);
16     int T;
17     cin >> T;
18     for(int t_i = 0; t_i < T; t_i++)
19     {
20         int N;
21         cin >> N;
22         int X;
23         cin >> X;
24         vector<int> A(N);
```

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Test against custom input ▼

Custom input populated ↺

Compile & Test code

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Output Format

For each test case in a new line, print an integer representing the maximum friends she can invite.

Constraints

$$1 \leq T \leq 10$$

$$1 \leq N \leq 10^4$$

$$1 \leq X \leq 10^4$$

$$1 \leq A[i] \leq 10^9$$

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input



Sample output



```
1
5
4
2 3 5 2 3
```

```
3
```

New Submission

All Submissions

❶ Auto-complete connection failed.
Reconnect

Save

C++17 (g++ 10.3.0)



```
1 #include<bits/stdc++.h>
2 using namespace std;
3
4 long long solve (int N, int X, vector<int> A) {
5     // code starts here
6     sort(A.begin(),A.end());
7     long long ans = 0;
8     priority_queue<int,vector<int>,greater<int>>pq;
9     for(int )
10 }
11
12 int main() {
13
14     ios::sync_with_stdio(0);
15     cin.tie(0);
16     int T;
17     cin >> T;
18     for(int t_i = 0; t_i < T; t_i++)
19     {
20         int N;
21         cin >> N;
22         int X;
23         cin >> X;
24         vector<int> A(N);
```

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Test against custom input

Custom input populated

Compile & Test code

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Example

Assumptions

- $T = 1$
- $N = 4$
- $A = "bbac"$
- $D = 1$

Approach

- You have $f(a) = 97, f(b) = 98, f(c) = 99$.
- For first character, you have $i = 1$, possible values of $j > i$ such that $|f(A_i) - f(A_j)| \leq 1$ are $j = 2, 3, 4$. Therefore, 3 such indices exist.
- For second character, you have $i = 2$, possible values of $j > i$ such that $|f(A_i) - f(A_j)| \leq 1$ are $j = 3, 4$. Therefore, 2 such indices exist.
- For third character, you have $i = 3$, there is no possible value of $j > i$ such that $|f(A_i) - f(A_j)| \leq 1$. Therefore, 0 such indices exist.
- For fourth character, you have $i = 4$, there is no possible value of $j > i$ such that $|f(A_i) - f(A_j)| \leq 1$. Therefore, 0 such indices exist.
- Therefore, the output is 3 2 0 0.

Function Description

New Submission

All Submissions

Auto-complete ready!

Save

C (gcc 10.3)



```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int* d_distance (int N, char* A, int D, int* out_n) {
6     // Write your code here
7
8 }
9
10 int main() {
11     int out_n;
12     int T;
13     scanf("%d", &T);
14     for(int t_i = 0; t_i < T; t_i++)
15     {
16         int N;
17         scanf("%d", &N);
18         char* A = (char*)malloc((N+1) * sizeof(char));
19         scanf("\n%[^\n]s", A);
20         int D;
21         scanf("%d", &D);
22
23         int* out_ = d_distance(N, A, D, &out_n);
24         printf("%d", out_[0]);
```

1:1 vscode



Test against custom input

Custom input populated

Compile & Test code

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Function Description

Complete the `d_distance` function provided in the editor. This function takes the following 3 parameters and returns an array/vector of integers of length N whose i^{th} integer denotes the count of indices $j > i$ such that $|f(A_i) - f(A_j)| \leq D$:

- N : Represents the integer N
- A : Represents the string A
- D : Represent the integer D

Input format

Note: This is the input format that you must use to provide custom input (available above the Compile and Test button).

- The first line contains an integer T denoting the number of test cases. T also specifies the number of times you have to run the `d_distance` function on a different set of inputs.
- For each test case:
 - The first line contains an integer N denoting the length of string A .
 - The second line contains a string A of length N .
 - The third line contains an integer D .

Output format

For each test case in a separate line, print N space-separated

New Submission

All Submissions

Auto-complete ready!

Save

C (gcc 10.3)



```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int* d_distance (int N, char* A, int D, int* out_n) {
6     // Write your code here
7
8 }
9
10 int main() {
11     int out_n;
12     int T;
13     scanf("%d", &T);
14     for(int t_i = 0; t_i < T; t_i++)
15     {
16         int N;
17         scanf("%d", &N);
18         char* A = (char*)malloc((N+1) * sizeof(char));
19         scanf("\n%[^\n]s", A);
20         int D;
21         scanf("%d", &D);
22
23         int* out_ = d_distance(N, A, D, &out_n);
24         printf("%d", out_[0]);
```

1:1 vscode



Test against custom input

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Compile & Test code

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**Output format**

For each test case in a separate line, print N space-separated integers, where the i^{th} integer denotes the count of indices $j > i$ such that $|f(A_i) - f(A_j)| \leq D$

1

2

Constraints

$$1 \leq T \leq 10$$

$$1 \leq N \leq 2 \times 10^5$$

String A consists of lower case English alphabets.

$$0 \leq D \leq 25$$

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input



Sample output



```
2
4
bbac
1
2
aa
1
```

```
3 2 0 0
1 0
```

New Submission

All Submissions

Auto-complete ready!

Save

C (gcc 10.3)



```
1 #include<stdio.h>
2 #include<stdbool.h>
3 #include<malloc.h>
4
5 int* d_distance (int N, char* A, int D, int* out_n) {
6     // Write your code here
7
8 }
9
10 int main() {
11     int out_n;
12     int T;
13     scanf("%d", &T);
14     for(int t_i = 0; t_i < T; t_i++)
15     {
16         int N;
17         scanf("%d", &N);
18         char* A = (char*)malloc((N+1) * sizeof(char));
19         scanf("\n%[^\n]s", A);
20         int D;
21         scanf("%d", &D);
22
23         int* out_ = d_distance(N, A, D, &out_n);
24         printf("%d", out_[0]);
```

1:1 vscode



Test against custom input

Custom input populated

Compile & Test code

Submit code

- You have $f(a) = 97$
- For first character, you have $i = 1$, possible values of $j > i$ such that $|f(A_i) - f(A_j)| \leq 0$ are $j = 2$
- For third character, you have $i = 2$, there is no possible value of $j > i$ such that $|f(A_i) - f(A_j)| \leq 0$.
- Therefore, the output is 10

The following question has been asked

Sample input 1

```
10
9
jbiجهjce
1
7
dacgkge
1
9
fcgaceifk
4
9
eeecbfhgd
1
9
fcgaceifk
4
```

[View more](#)

Sample input 2

```
10
29
ttjamliissbrgpiocojinqjl
7
```

Sample output 2

```
11 10 18 6 18 18 16 9 8 9 7
10 9 6 6 5 8 9 9 7 6 3 2 0 2
10 16 9 16 11 13 12 12 10 9
12 18 13 10 14 6 8 6 7 8 4 3
```

New Submission

All Submissions

Auto-complete ready!

Save

C (gcc 10.3)



```
1 #include<stdio.h>
2 #include<stdlib.h>
```

Sample input 1

```
10
9
jbiجهjce
1
7
dacgkge
1
9
fcgaceifk
4
9
eeecbfhgd
1
9
fcgaceifk
4
ekhgjcih
```

Sample output 1

```
3 1 3 1 1 0 0 0 0
2 0 0 1 0 0 0
6 5 5 2 2 2 2 0 0
4 3 2 2 0 1 1 0 0
4 4 4 3 2 0 1 0
2 1 0 0 0
5 3 2 3 2 1 0
3 3 1 1 0
0 2 1 0 0 0 1 0 0
2 3 1 0 0
```

24

```
printf("%d", out[0]);
```

1:1 vscode



Test against custom input

Custom Input populated

Compile & Test code

Submit code

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1

2

- The first line contains T denoting the number of test cases. T also specifies the number of times you have to run the `solve` function on a different set of inputs.
- For each test case:
 - The first line contains an integer N which represents the size of the array.
 - The second line contains the integer X
 - The third line contains N space-separated integers denoting the number of candies of type i available to her.

Output Format

For each test case in a new line, print an integer representing the maximum friends she can invite.

Constraints

$$1 \leq T \leq 10$$

$$1 \leq N \leq 10^4$$

$$1 \leq X \leq 10^4$$

$$1 \leq A[i] \leq 10^9$$

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and

```
4 long long solve (int N, int X, vector<int> A) {
5     // code starts here
6     priority_queue<int> pq;
7     for(int i:A){
8         if(i > 0){
9             pq.push(i);
10        }
11    }
12    long long ans = 0;
13    vector<int> cache;
14    while(pq.size() >= X){
15        cache.clear();
16        int Y = X;
17        while(Y--){
18            if(pq.top() > 1){
19                cache.push_back(pq.top() - 1);
20            }
21            pq.pop();
22        }
23        for(int j:cache){
24            pq.push(j);
25        }
26        ans++;
27    }
```

21:21 vscode



Test against custom input ▼

Custom input populated ↺

Compile & Test code

Submit code

Submission ID: 76921093 / 1 minute ago

RESULT: Partially accepted

[Refer Judge environment](#)

Score

Time (sec)

Memory (KiB)

Language